Luci Autonomous Wheelchair

1 Important Information

1.1 Get to know the system

The Luci wheelchair is an autonomous wheelchair system that is attached to common wheelchairs. The one that we have in our lab is the Quickie Q500m wheelchair. The spec sheet for the wheelchair is located in the lab in a box titled, Luci wheelchair. There you will see a manual for the wheelchair as well as a manual for the Luci system.

1.2 Powering the system

Figure 1: The main controller for the wheelchair. Contains the power switch, joystick, and Luci override button

In order to power on the wheelchair, you must move the lever on the left side of the wheelchair, with the green power button, up. Once you do this the wheelchair will turn on, and after a few seconds will be ready to use as Luci intended. It will be controllable from the hand joystick and will have collision avoidance enabled. To turn off the collision avoidance, just press the Luci button on the top panel on the left arm rest. This will put the Luci chair into an override mode with a reduced speed. This mode will last for 30 seconds, or until turned off. The collision mode is also set to bring the chair to a complete stop if it gets too close to a sensed danger. The duration of the override mode and the collision mode can be changed in the setup screen, which is discussed in a later section.

1.3 Maintenence and Updates

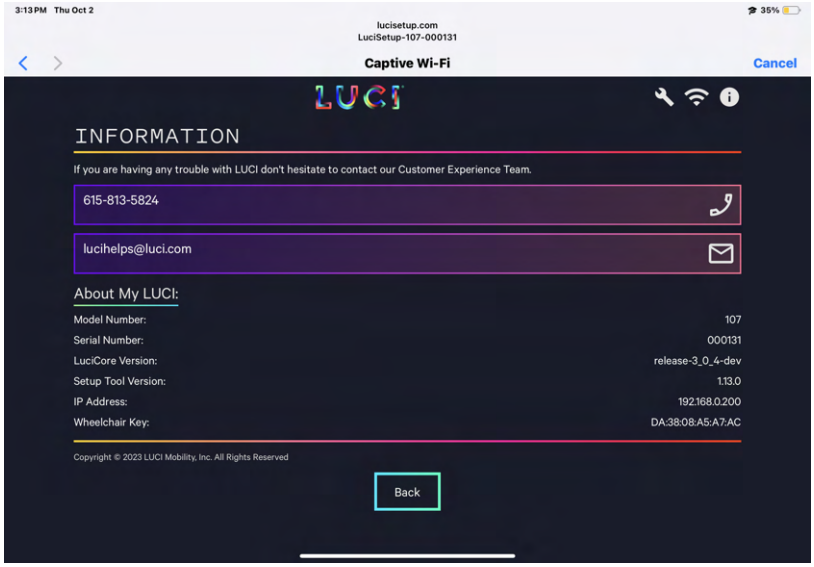

Figure 2: The setup screen for how to check the software version that the luci system is on

To access the underlying Luci system, press and hold the Luci button until a sound is played. Then connect to the chair's wifi on any device, which will give you access to the setup menu. Within this setup menu you will be able to change things like: the chair's connected wifi, response speed for the joystick, and check for software version that is running on the chair. Additionally, this is how you can access software versions on the wheelchair. Most of the system updates are done by the Luci team over wifi, so as long as the wheelchair has power it should be able to be updated by the Luci team. If you are unclear on if the chair is up to date, and the check for updates button isn't responding, email Josh our contact at Luci for the wheelchair.

2 Getting Started

There are several different nodes setup with the chair to work with ROS2, as well as a docker image that is created for the wheelchair to work with the pupil labs neon ROS2 wrapper. Currently the chair has a github repository for a couple topics, with those being: transforms, grpc, msgs, bringup, teleop, odom, slam, and the SDK.

2.1 Luci Developed Packages

Luci has developed a lot of nodes for this wheelchair, while also having their own docs for reference. Within their docs, they refer to their github, which contains their docker image, transforms repo, grpc repo, and other various repos that correlate to the wheelchair. For the most part, the main repos that we need from the Luci repo are the grpc and transforms. Which contains the package for communicating to the wheelchair, and the package for the transforms for our specific robot. The next packages are the ones created and managed by out lab.

2.2 Bringup

The bringup node contains a launch file for getting the wheelchair connected and broadcasting data within the lab. It is set to the wheelchairs pre-established IP for the lab's wifi. In order to see what the chair's IP is and how to change it, refer to section 1.3, which references how to enter setup. This data can be visualized in rviz2 in which you will be able to see a few pointclouds that are generated by the wheelchair, with the most visual being the camera points pointcloud. The github mentioned earlier also contains a set by step guide on how to install this package.

2.3 Teleop

The teleop node contains the necessary for controlling the Luci wheelchair with teleop. It can be used to control the wheelchair with a keyboard and with an xbox controller. For emergency stops when in teleop, you have a couple of options. For emergency stop with a person sitting in the wheelchair, moving the joystick on the wheelchair causes all controls being sent to the wheelchair to stop. Additionally it puts the chair in idle mode so nothing can control it until something else claims control, whether that be moving the joystick again, or pressing the control button on the controller. For the stop with the controller, pressing the B button causes the wheelchair to enter the same standby mode as with the joystick , and remains that way until the control is claimed again.

2.4 Odom

The odom is currently a work in progress, but as of its current state, it takes the encoder values from the the `\luci\encoders` topic and calculates wheel angle, distance traveled, position, and velocity.

2.5 Slam

The slam utilizes the ROS2 slam_toolbox to generate a map of the wheelchairs environments. It works by converting the camera pointcloud to laserscan, then combining that with the wheelchair's odom and imu data to create the map. Currently a work in progress as the slam maps that are being created aren't accurate.

2.6 Safety

The Quickie Q500M wheelchair includes several built-in safety features, primarily through the LUCI system, which helps prevent collisions and enhances control during navigation. The LUCI system automatically performs speed adjustments and emergency stops when necessary to maintain safe operation. When interacting with the wheelchair, avoid placing hands, feet, or other body parts under the seat, near the wheels, or around the base of the wheelchair. These areas contain moving components and the high-voltage battery. The armrests of the chair are adjustable; however, use caution when moving the controller arm, as it contains wiring critical to both chair movement and LUCI system functionality.

To charge the wheelchair, use the Quickie Q500M/Q400M Series Battery Charger. The charging port is located on the arm of the chair that houses the joystick controller. Charging typically takes 6–12 hours, depending on the battery level, and overnight charging is recommended. The wheelchair can safely remain connected to a Sunrise Medical approved charger for up to 24 hours without damaging the battery. For users who operate the chair daily, regular overnight charging is recommended. If the chair will remain plugged in for more than 26 hours, disconnect it from the charger at the wheelchair connection point and unplug the charger from the wall.

2.7 More Information

For more information please look at a few of the other docs in the google drive.